Practical 2

Ex.1 DVDs Checked Out

x (No. of DVDs)	p(x)	x · p(x)	x^2	$x^2 \cdot p(x)$
1	0.30	0.3000	1	0.3000
2	0.25	0.5000	4	1.0000
3	0.15	0.4500	9	1.3500
4	0.05	0.2000	16	0.8000
5	0.10	0.5000	25	2.5000
6	0.15	0.9000	36	5.4000
Total	**1.00**	**2.8500**		**11.3500**

*Find the mean, variance and SD	Symbol	value	Equation	Answer
[1] Mean of X	E(X)	2.8500	$E(X) = \Sigma\, x \cdot p(x)$	**2.8500**
[2] Variance of X	Var(X)	11.3500	$Var(X) = E(X^2) - [E(X)]^2$	**3.2275**
[3] Standard deviation of X	σ	3.2275	$\sigma = \sqrt{Var(X)}$	**1.7965**

Ex.2 Balanced Die

x (points rolled)	p(x)	x · p(x)	x^2	$x^2 \cdot p(x)$
1	0.1667	0.1667	1	0.1667
2	0.1667	0.3333	4	0.6667
3	0.1667	0.5000	9	1.5000
4	0.1667	0.6667	16	2.6667
5	0.1667	0.8333	25	4.1667
6	0.1667	1.0000	36	6.0000
Total	**1.0000**	**3.5000**		**15.1667**

*Find the variance	Symbol	value	Equation	Answer
[1] Mean of X	E(X)	3.5000	$E(X) = \Sigma\, x \cdot p(x)$	**3.5000**
[2] Variance of X	Var(X)	15.1667	$Var(X) = E(X^2) - [E(X)]^2$	**2.9167**

Ex.3 Number of Seizures

x (seizures per year)	p(x)	x · p(x)	x^2	$x^2 \cdot p(x)$
0	0.17	0.0000	0	0.0000
2	0.21	0.4200	4	0.8400
4	0.18	0.7200	16	2.8800
6	0.11	0.6600	36	3.9600
8	0.16	1.2800	64	10.2400
10	0.17	1.7000	100	17.0000
Total	**1.00**	**4.7800**		**34.9200**

Constants in part (c): $E(aX^2 + bX + c)$	Value
a	200
b	1
c	-5

*Find the following	Symbol	value	Equation	Answer
(a) Expected number of seizures	E(X)	4.7800	$E(X) = \Sigma\, x \cdot p(x)$	**4.7800**
(b) $E(X^2)$	$E(X^2)$	34.9200	$E(X^2) = \Sigma\, x^2 \cdot p(x)$	**34.9200**
(c) $E(200X^2 + X - 5)$	$E(aX^2+bX+c)$		$E(aX^2+bX+c) = a \cdot E(X^2) + b \cdot E(X) + c$	**6983.7800**
(d) Variance of X	Var(X)	34.9200	$Var(X) = E(X^2) - [E(X)]^2$	**12.0716**

Ex.4 Continuous Random Variable f(x) = 2e^(−2x), x > 0

Items	Value
Rate λ (f(x) = λ e^(−λx) for x > 0, else 0)	2

*Find the mean and variance	Symbol	value	Equation	Answer
[1] Mean of X	E(X)	2.0000	$E(X) = \int_{[0,\infty]} x \cdot \lambda e^{(-\lambda x)}\, dx = 1/\lambda$	**0.5000**
[2] Second moment	$E(X^2)$	2.0000	$E(X^2) = \int_{[0,\infty]} x^2 \cdot \lambda e^{(-\lambda x)}\, dx = 2/\lambda^2$	**0.5000**
[3] Variance of X	Var(X)	0.5000	$Var(X) = E(X^2) - [E(X)]^2$	**0.2500**

Ex.5 Laptop Service Life f(x) = (1/4.5) e^(−x/4.5), x > 0

Items	Value
θ (f(x) = (1/θ) e^(−x/θ) for x > 0, else 0)	4.5
Rate λ = 1/θ	0.2222

*Find the mean	Symbol	value	Equation	Answer
[1] Mean service life (years)	E(X)	0.2222	$E(X) = \int_{[0,\infty]} x \cdot \lambda e^{(-\lambda x)}\, dx = 1/\lambda$	**4.5000**